

Multiple Choice

1. **Answer:** D

Options: A) Melting point; B) Boiling point; C) Polarizability; D) First ionization energy

Note: Correct option: D - First ionization energy

2. **Answer:** D

Options: A) 10.8×10^{-5} ; B) 4.11×10^{-5} ; C) 3.43×10^{-5} ; D) 1.71×10^{-5}

Note: Correct option : D – 1.71×10^{-5}

2. **Answer:** D

Options: A) Q = 1012; B) Q = 2012; C) Q = 3012; D) Q = 6012

Note: Correct option: D - Q = 6012

3. **Answer:** A

Options: A) 3; B) 5; C) 8; D) 4

Note: Correct option: A - 3

4. **Answer:** B

Options: A) The I⁻ : IO₃⁻ ratio is 3:1.; B) The MnO₄⁻ : I⁻ ratio is 6:5.; C) The MnO₄⁻ : Mn₂⁺ ratio is 3:1.; D) The H⁺ : I⁻ ratio is 2:1.

Note: Correct option: B - The MnO₄⁻ : I⁻ ratio is 6:5.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Raman'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It has little effect.'

Long Form Response

11. **Answer:** T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.050 M $AlCl_3$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key